Status	Finished
Started	Monday, 12 October 2026, 5:26 AM
Completed	Monday, 12 October 2026, 5:30 AM
Duration	4 mins 6 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. At the beginning of the list
b. In the middle of the list
c. In the right sublist ✓
d. In the left sublist

Question **2**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99
b. 90 and 99 ✓
c. 89 and 94
d. 90 and 94

Question **3**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1
b. n/2
c. log n
d. n ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Binary search
b. Hashing
c. Jump search
d. Linear search ✓

Question **5**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By dividing the list length by two ✓
b. By starting from the first element
c. By comparing each element sequentially
d. By using a hash function

Question **6**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The first element is returned
c. The last element is returned
d. The search is considered unsuccessful ✓

Question **7**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list
b. It can be applied only if the list is sorted ✓
c. It compares elements sequentially
d. It works on unsorted lists

Question **8**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the search algorithm finishes
b. When the list contains duplicate elements
c. When the list is sorted
d. When the element is found in the list ✓

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Skipping every second element
b. Sorting the list before searching
c. Dividing the list in half repeatedly
d. Checking each element sequentially ✓

Question **10**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is the last element in the array
b. When the item is somewhere in the middle of the array ✓
c. When the item is not the array at all
d. Item is the last element in the array or item is not there at all

Question **11**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Binary search
b. Interpolation search
c. Hash search
d. Linear search ✓

Question **12**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By sorting the list first
b. By using a hash function
c. By comparing each element in the list sequentially ✓
d. By dividing the list into halves

Question **13**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n
b. n/2
c. 1
d. n ✓

Question **14**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Complexity

b. Bubble ✓

c. Insertion

d. Selection

Question **15**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Binary search

b. Both (1) & (3)

c. Hash search

d. Linear search ✓